

Rational Numbers on the Number Line

Lesson 9-4

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

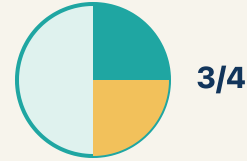
Picture first, then the word, then a plain-language meaning. Say each word out loud.



$\frac{1}{2} = 0.5$, $-\frac{3}{4} = -0.75$, $6 = \frac{6}{1}$ — all rational numbers

Rational number

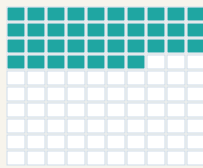
Write the definition:



$\frac{3}{4}$ means 3 out of 4 equal parts — like 3 slices of a pizza cut into 4

Fraction

Write the definition:



0.37

$0.5 = \frac{1}{2}$ (halfway between 0 and 1 on the number line)

Decimal

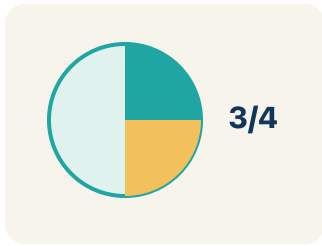
Write the definition:



A line with marks at -2, -1.5, -1, -0.5, 0, 0.5, 1, 1.5, 2

Number line

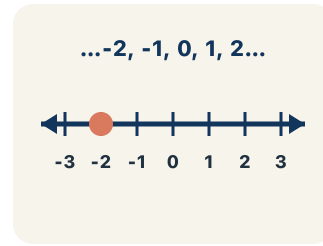
Write the definition:



$\frac{1}{2} = 0.5 = \frac{2}{4}$ — all name the same point on the number line

Equivalent

Write the definition:



..., -3, -2, -1, 0, 1, 2, 3, ...

Integer

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can place rational numbers, including fractions and decimals, on a number line.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Rational number

Fraction

Decimal

Number line

Equivalent

Integer

1 Any number that can be written as a fraction, including integers and decimals, is a _____.

2 A number that shows part of a whole, like $\frac{3}{5}$, is a _____.

3 A number that uses a point to show parts of a whole is a _____.

4 A line with numbers in order used to place and compare values is a _____.

5 Numbers that name the same value, like $\frac{1}{2}$ and 0.5, are _____.

6 A whole number or its opposite is an _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: Which point is located between -1 and -2 on the number line?

- A. -0.5
- B. -1.5
- C. -2.5
- D. 1.5

Step 1 -1.5 is between -1 and -2 because it is 0.5 units to the left of -1 (or 0.5 units to the right of -2).

 **Answer:** B. -1.5

 We do — together

Solve this one with your class using the same steps.

Problem: Which fraction is equivalent to 0.25?

- A. $\frac{1}{4}$
- B. $\frac{1}{2}$
- C. $\frac{2}{5}$
- D. $\frac{1}{3}$

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: Where is $-\frac{3}{4}$ located on the number line?

- A. Between 0 and -1
- B. Between -3 and -4
- C. Between -1 and -2
- D. Between 0 and 1

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Order these three trail markers from least to greatest: -1.5, 0.5, -0.5.

- A. -1.5, -0.5, 0.5
- B. 0.5, -0.5, -1.5
- C. -0.5, -1.5, 0.5
- D. -1.5, 0.5, -0.5

Show your work:

2. On the trail map, $-\frac{3}{4}$ sits to the LEFT of which marker?

- A. $-\frac{1}{4}$
- B. -1
- C. -2
- D. $-\frac{7}{8}$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Name three different rational numbers between -1 and 0. Write each as both a fraction and a decimal. Explain how you know they are between -1 and 0.

Sentence starter: Three rational numbers between -1 and 0 are: $\frac{\quad}{\quad} = \quad$, $\frac{\quad}{\quad} = \quad$, and $\frac{\quad}{\quad} = \quad$. I know they are between -1 and 0 because $\quad$.

Show your work:

Reflect — Exit Ticket

Which rational number is located between -2 and -3 on the number line?

- A. -1.5
- B. -2.75
- C. -3.5
- D. 2.5

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Rational number
2. **Notes 2:** Fraction
3. **Notes 3:** Decimal
4. **Notes 4:** Number line
5. **Notes 5:** Equivalent
6. **Notes 6:** Integer
7. **We Try (notes):** A. $\frac{1}{4} - 0.25$ means 25 hundredths = $\frac{25}{100} = \frac{1}{4}$.
8. **You Try (notes):** A. Between 0 and -1 — $-\frac{3}{4} = -0.75$, which is between 0 and -1. It is $\frac{3}{4}$ of the way from 0 toward -1.
9. **Try It 1:** A. -1.5, -0.5, 0.5 — The farthest left on the number line comes first: -1.5, then -0.5, then 0.5.
10. **Try It 2:** A. $-\frac{1}{4} - \frac{3}{4} = -0.75$ and $-\frac{1}{4} = -0.25$. Since -0.75 is farther left (less), $-\frac{3}{4}$ is to the left of $-\frac{1}{4}$.
11. **Exit Ticket:** B. -2.75 — -2.75 is between -2 and -3 because it is 0.75 units to the left of -2. -1.5 is between -1 and -2, and -3.5 is between -3 and -4.